

Cut elimination for G3cp

Theorem 3.2.3: the rule of cut:

$$\frac{\Gamma \Rightarrow \Delta, D \quad D, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} \text{cut}$$

is admissible in G3cp.

The proof will be by double induction: main induction on the weight of the cut formula, subinduction on the cut-height of the cut application. w stands for $w(D)$ and h for the cut-height.

G3cp

Logical axiom:

$$P, \Gamma \Rightarrow \Delta, P$$

Logical rules:

$$\frac{A, B, \Gamma \Rightarrow \Delta}{A \& B, \Gamma \Rightarrow \Delta} L\&$$

$$\frac{\Gamma \Rightarrow \Delta, A \quad \Gamma \Rightarrow \Delta, B}{\Gamma \Rightarrow \Delta, A \& B} R\&$$

$$\frac{A, \Gamma \Rightarrow \Delta \quad B, \Gamma \Rightarrow \Delta}{A \vee B, \Gamma \Rightarrow \Delta} L\vee$$

$$\frac{\Gamma \Rightarrow \Delta, A, B}{\Gamma \Rightarrow \Delta, A \vee B} R\vee$$

$$\frac{\Gamma \Rightarrow \Delta, A \quad B, \Gamma \Rightarrow \Delta}{A \supset B, \Gamma \Rightarrow \Delta} L\supset$$

$$\frac{A, \Gamma \Rightarrow \Delta, B}{\Gamma \Rightarrow \Delta, A \supset B} R\supset$$

$$\frac{}{\perp, \Gamma \Rightarrow \Delta} L\perp$$

In this system weakening and contraction are (height-preservingly) admissible. The height preservation is not needed in this proof.

Definitions:

Weight is defined inductively:

$$w(\perp) = 0;$$

$$w(P) = 1 \text{ for an atomic formula } P;$$

$$w(A \circ B) = w(A) + w(B) + 1.$$

The **height** of a derivation is the greatest number of successive applications of rules in it. The conclusion of $L\perp$, and an axiom, have height 0.

Cut-height is defined as the sum of the heights of the cut's premises.

The formula with the connective in a rule is the **principal formula** of that rule, and its components in the premises are the **active formulas**.

The rest of the terms are used in a usual manner.

Proof:

We first prove that for $w = 0$, and any h , cut is admissible.

Subinduction base:

For the base case $w = 0, h = 0$:

$$\frac{\Gamma \Rightarrow \Delta, \perp \quad \perp, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} cut$$

Since the cut-height is 0, the left premise is an axiom or conclusion of $L\perp$, and so is $\Gamma, \Pi \Rightarrow \Delta, \Lambda$ by observation.

Subinduction step:

For all n , if cut is admissible when $w = 0$ and $h = i$, for all $i \leq n$, then cut is admissible when $w = 0$ and $h = n + 1$.

Suppose a derivation of cut-height $n + 1$:

$$\frac{\Gamma \Rightarrow \Delta, \perp \quad \perp, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} cut$$

Note that the right premise being an instance of $L\perp$ would not help here. Since $\perp$ cannot be principal in the left premise, we will consider how the left premise has been derived in the subcases below:

1. $L\wedge$:

$$\frac{\frac{\Gamma, A, B \Rightarrow \Delta, \perp}{\Gamma, A \wedge B \Rightarrow \Delta, \perp} L\wedge \quad \perp, \Pi \Rightarrow \Lambda}{\Gamma, A \wedge B, \Pi \Rightarrow \Delta, \Lambda} cut$$

cut-height is $n + 1 + m$.^[1] The contexts would not be marked with $'$, but it is implicit.

We can push the cut upward:

$$\frac{\frac{\Gamma, A, B \Rightarrow \Delta, \perp \quad \perp, \Pi \Rightarrow \Lambda}{\Gamma, A, B, \Pi \Rightarrow \Delta, \Lambda} cut}{\Gamma, A \wedge B, \Pi \Rightarrow \Delta, \Lambda} L\wedge$$

cut-height is now $n + m$, which is one less. By the subinduction hypothesis, the premise of $L\wedge$ is derivable (since the cut premises are too) and therefore so is the final conclusion by applying the rule.

2. $L\vee$:

$$\frac{\frac{A, \Gamma \Rightarrow \Delta, \perp \quad B, \Gamma \Rightarrow \Delta, \perp}{A \vee B, \Gamma \Rightarrow \Delta, \perp} L\vee \quad \perp, \Pi \Rightarrow \Lambda}{A \vee B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} cut$$

cut-height is $\max(n, m) + 1 + k$

We can push the cut upward:

$$\frac{\frac{A, \Gamma \Rightarrow \Delta, \perp \quad \perp, \Pi \Rightarrow \Lambda}{A, \Gamma, \Pi \Rightarrow \Delta, \Lambda} cut \quad \frac{B, \Gamma \Rightarrow \Delta, \perp \quad \perp, \Pi \Rightarrow \Lambda}{B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} cut}{A \vee B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} L\vee$$

Now the cut-height for the left cut is $n + k$ and for the right cut $m + k$, each of which is strictly less than $\max(n, m) + 1 + k$. By the subinduction hypothesis, both premises of $L\vee$ are derivable and therefore so is the conclusion.

3. $L\rightarrow$:

$$\frac{\frac{\Gamma \Rightarrow \Delta, \perp, A \quad B, \Gamma \Rightarrow \Delta, \perp}{A \rightarrow B, \Gamma \Rightarrow \Delta, \perp} L\rightarrow \quad \perp, \Pi \Rightarrow \Lambda}{A \rightarrow B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} cut$$

cut-height is $\max(n, m) + 1 + k$

We push the cut upward in a similar fashion to (2):

$$\frac{\frac{\Gamma \Rightarrow \Delta, A, \perp \quad \perp, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, \Lambda, A} cut \quad \frac{B, \Gamma \Rightarrow \Delta, \perp \quad \perp, \Pi \Rightarrow \Lambda}{B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} cut}{A \rightarrow B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} L\rightarrow$$

As (2), left cut $n + k$ and right cut $m + k$, strictly less. By the subinduction hypothesis and rule application the conclusion is derivable.

4. $L\perp$ or axiom:

Even though the right premise is an instance of $L\perp$ it might be a derivation of a longer height (last rule not being $L\perp$), contributing to the cut-height, while the left premise could be an axiom or the result of applying $L\perp$. In such a case the conclusion would be too and therefore the conclusion is derivable without applying cut.

The $R\wedge$ is symmetrical to $L\vee$, and $R\vee$ to $L\wedge$.

5. $R\rightarrow$:

$$\frac{\frac{A, \Gamma \Rightarrow \Delta, B, \perp}{\Gamma \Rightarrow \Delta, A \rightarrow B, \perp} R \rightarrow \quad \perp, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, A \rightarrow B, \Lambda} cut$$

Cut-height is $n + 1 + m$. We can push the cut upward:

$$\frac{\frac{A, \Gamma \Rightarrow \Delta, B, \perp \quad \perp, \Pi \Rightarrow \Lambda}{A, \Gamma, \Pi \Rightarrow \Delta, B, \Lambda} cut}{\Gamma, \Pi \Rightarrow \Delta, A \rightarrow B, \Lambda} R \rightarrow$$

Cut-height is now $n + m$, which is strictly less. By the subinduction hypothesis and applying the rule the conclusion is derivable.

The cases above will be used as a template later for cases (i) and (ii) under the subinduction step for the main induction step.

Therefore, for $w = 0$, and any h , cut is admissible.

Now we will do the main induction step.

Main induction step:

For all k , if cut is admissible when $w = i$, $h = n$ for all $i \leq k$ and for all $n \in \mathbb{N}$, then cut is admissible when $w = k + 1$, $h = n$ for all $n \in \mathbb{N}$.

This will be proven using subinduction on cut-height.

Subinduction base:

Cut is admissible when $w = k + 1$ and $h = 0$.

We will start with a derivation in which $w(D) = k + 1$. In:

$$\frac{\Gamma \Rightarrow \Delta, D \quad D, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} cut$$

D has a non-zero weight and each of the two premises has a height of zero.

If the left premise is an instance of $L\perp$ then so is the conclusion. If it is an axiom, $P \in \Gamma, P \in \Delta$, then so is the conclusion. However, if it is an axiom but $D = P, P \in \Gamma$, then the right premise is either an instance of $L\perp$ in which case the conclusion would be too, or it is an axiom:

- i. $Q \in \Pi, Q \in \Lambda$;
- ii. $Q = D, Q \in \Lambda$

If (i) the conclusion would be axiom too. If (ii) then $P = Q = D$ and the conclusion would be an axiom. The base case for the subinduction is done.

Subinduction step:

For all m , if cut is admissible when $w = k + 1$, $h = i$ for all $i \leq m$, then cut is admissible when

$$w = k + 1, h = m + 1.$$

We will start with a derivation in which $w(D) = k + 1, h = m + 1$:

$$\frac{\Gamma \Rightarrow \Delta, D \quad D, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} \text{cut}$$

D has a non-zero weight, and at least one of the premises has a non-zero cut-height. The subcases will be with respect to D .

i. $D = P$:

Then the previous cases (1-5) apply similarly except in case (4) where the left premise is an axiom $P \in \Gamma$. In such a case, the left premise is $P, \Gamma' \Rightarrow \Delta, P$ and the conclusion of cut is $P, \Gamma', \Pi \Rightarrow \Delta, \Lambda$. We weaken the right premise to obtain $P, \Gamma', \Pi \Rightarrow \Delta, \Lambda$ which is the cut's conclusion. This does not need any appeal to the induction hypotheses or the height-preserving nature of weakening's admissibility, only that weakening is admissible.

ii. $D = A \circ B$:

If the left premise is initial then by observing that D is compound, the conclusion would be initial, the same argument applies if instead the right premise is initial. If D is not principal in the rule application used to derive the left premise, then the previous cases (1-5) also apply similarly. However, if it is principal in the left premise only, we will consider the right premise's last rule application, and the cases are symmetrical to the previous ones mentioned above. The subcases are exhausted.

If D is principal in both premises:

1. $D = A \wedge B$:

$$\frac{\frac{\Gamma \Rightarrow \Delta, A \quad \Gamma \Rightarrow \Delta, B}{\Gamma \Rightarrow \Delta, A \wedge B} R\wedge \quad \frac{A, B, \Pi \Rightarrow \Lambda}{A \wedge B, \Pi \Rightarrow \Lambda} L\wedge}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} \text{cut}$$

The cut-height is $(\max(n, m) + 1) + (k + 1)$

It can be transformed into:

$$\frac{\Gamma \Rightarrow \Delta, B \quad \frac{\Gamma \Rightarrow \Delta, A \quad A, B, \Pi \Rightarrow \Lambda}{B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} \text{cut}}{\Gamma, \Gamma, \Pi \Rightarrow \Delta, \Delta, \Lambda} \text{ctr} \\ \Gamma, \Pi \Rightarrow \Delta, \Lambda$$

Contraction is applied (left or right) as many times as needed to reach the conclusion. The first cut's cut-height is $n + k$ which is strictly less, but we cannot apply the subinduction hypothesis due to the weight being different (reduced). Therefore its conclusion is derivable by the main

induction hypothesis. For the second cut, the cut-height is unknown but the weight is also reduced, so by the main hypothesis the conclusion is derivable.

2. $D = A \vee B$:

$$\frac{\frac{\Gamma \Rightarrow \Delta, A, B}{\Gamma \Rightarrow \Delta, A \vee B} R\vee \quad \frac{\frac{A, \Pi \Rightarrow \Lambda \quad B, \Pi \Rightarrow \Lambda}{A \vee B, \Pi \Rightarrow \Lambda} L\vee}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} cut$$

The cut-height is $(n + 1) + (\max(k, m) + 1)$. It can be transformed to:

$$\frac{\frac{\frac{\Gamma \Rightarrow \Delta, A, B \quad B, \Pi \Rightarrow \Lambda}{\Gamma, \Pi \Rightarrow \Delta, \Lambda, A} cut \quad \frac{A, \Pi \Rightarrow \Lambda}{\Gamma, \Pi, \Pi \Rightarrow \Delta, \Lambda, \Lambda} cut}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} Ctr$$

Contraction is applied (left or right) as many times as needed to reach the conclusion. We may analyze the cut-height reductions but that is not relevant to the induction hypotheses since weight is reduced in both cuts. Given that in the first cut weight is reduced, the main induction hypothesis applies and its conclusion is derivable, the same argument goes for the second cut.

3. $D = A \rightarrow B$:

$$\frac{\frac{A, \Gamma \Rightarrow \Delta, B}{\Gamma \Rightarrow \Delta, A \rightarrow B} R \rightarrow \quad \frac{\frac{\Pi \Rightarrow \Lambda, A \quad B, \Pi \Rightarrow \Lambda}{A \rightarrow B, \Pi \Rightarrow \Lambda} L \rightarrow}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} cut$$

The cut-height is $(n + 1) + (\max(k, m) + 1)$. It can be transformed to:

$$\frac{\frac{\frac{\Pi \Rightarrow \Lambda, A \quad A, \Gamma \Rightarrow \Delta, B}{\Gamma, \Pi \Rightarrow \Delta, \Lambda, B} cut \quad \frac{B, \Pi \Rightarrow \Lambda}{\Gamma, \Pi, \Pi \Rightarrow \Delta, \Lambda, \Lambda} cut}{\Gamma, \Pi \Rightarrow \Delta, \Lambda} Ctr$$

Contraction is applied (left or right) as many times as needed to reach the conclusion. Weight is reduced for each cut, so by the main induction hypothesis each cut's conclusion is derivable.

The subinduction step is done, we thus proved cut is admissible when $w = k + 1$, and for any value of h .

The main induction step is done, we thus proved cut is admissible for any value of w and any value of h . $\square$

Notes:

Strictly speaking, the proof is that of cut admissibility rather than cut elimination, although they are equivalent.

Order of induction:

The main induction is on weight, rather than height, mainly because in the cases where D is principal in both premises, we guarantee a reduction in weight and sometimes a reduction in cut-height. The latter reduction is useless, since the subinduction hypothesis requires that the weight is fixed, and the main induction hypothesis only requires weight reduction so height is irrelevant. If the main induction were on height, the cut-height does not remain fixed so the subinduction hypothesis (which would be applicable when weight is reduced and height is fixed) is useless, but so is the main induction hypothesis since height reduction for all cuts is not guaranteed either. **An observation that can be made to summarize the failure:** in principal-principal cases, cut-height is uncontrollable but weight reduction is guaranteed.

What licenses the proof:

It becomes quickly noticeable that we can permute cut upward, or apply the cuts on formulas of less complexity, or both. What about our system's rules permits this? It is difficult to see why if the proof strategy is not attempted in other systems hostile to it, e.g., some modal logic systems. The main ingredients, at least here, are i) lack of context restrictions; ii) the active formulas have less complexity compared to the principal formula.

Compressing the proof:

This proof exposes the double induction structure, but the proof cases can be split like so:

Cut with an axiom or conclusion of $L\perp$ as at least one premise:

- i) Cut with an axiom or conclusion of $L\perp$ as the left premise.
- ii) Cut with an axiom or conclusion of $L\perp$ as the right premise.

Cut with neither premise an axiom or conclusion of $L\perp$:

- i) The cut formula is not principal in the left premise.
- ii) The cut formula is principal in the left premise only.
- iii) The cut formula is principal in both premises.

This exposes the procedural nature of cut elimination instead.

1. The naming convention for heights is: topsequents have heights (from left to right) $n, m, k, \dots$ ↩